

Offline Hindi Voice Assistant on ARM-Based SoC

PROJECT REPORT

Bharat AI SoC Student Challenge – PS-1

Submitted by:

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Regards

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ABSTRACT

This project presents the design and implementation of a fully offline Hindi voice assistant deployed on an ARM-based embedded platform (Raspberry Pi). The system captures audio input using a USB microphone, performs offline automatic speech recognition (ASR), recognizes predefined Hindi commands, and generates spoken responses using offline text-to-speech (TTS). The solution eliminates dependency on cloud services, ensuring data privacy, reduced latency, and reliable operation in low-connectivity environments. The assistant achieves response times under two seconds while operating within the computational constraints of embedded hardware.

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1. INTRODUCTION

Voice assistants have become very popular, but most of the existing solutions rely on cloud infrastructure for speech recognition and response generation. This approach has some drawbacks, including the requirement for constant internet connectivity, increased response latency, and possible privacy issues due to the transfer of voice data to cloud servers.

For this project, an offline Hindi voice assistant is created and implemented on a Raspberry Pi (ARM SoC). The voice assistant records audio from a USB microphone and recognizes speech using an offline Automatic Speech Recognition (ASR) engine. The recognized Hindi speech is then analyzed by a Python-based intent recognition module, which detects predefined commands and takes corresponding actions.

The voice assistant responds to the detected commands with audio generated by an offline Text-to-Speech (TTS) engine and plays it through a speaker, thus completing the entire process without the need for internet connectivity. The system is designed to handle multiple Hindi commands with low latency and efficient resource utilization, thus proving the viability of developing intelligent voice interfaces on resource-constrained embedded platforms.

2. PROBLEM STATEMENT

The aim of this project is to design and develop an entirely offline Hindi voice assistant running on an ARM-based embedded system. The system should be able to listen to voice commands, recognize Hindi speech, understand user commands, and speak responses offline without any internet connectivity.

The proposed solution must satisfy the following requirements:

- Operate completely without internet connectivity
- Accurately understand predefined Hindi voice commands
- Provide spoken responses with low latency (Response time less than 2)
- Run efficiently on ARM-based embedded hardware with limited computational resources

3. SYSTEM ARCHITECTURE

The system follows a local processing pipeline as shown below:

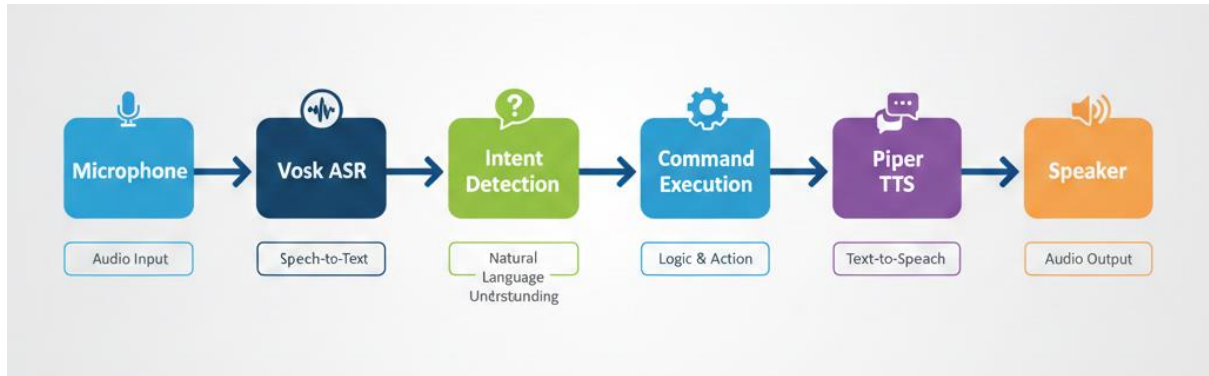


Figure 1: *Voice Assistant Functional Block Diagram.*

Audio Input

A USB microphone is used to capture live audio input from the user.

Automatic Speech Recognition (ASR)

Offline Hindi speech recognition is performed using the Vosk speech recognition engine. The ASR module converts speech signals into Hindi text locally on the device.

Intent Recognition

A Python-based keyword matching approach is used to recognize user intent from the transcribed Hindi text. This method is lightweight and suitable for real-time embedded systems.

Text-to-Speech (TTS)

Piper TTS is used for offline speech synthesis. The generated Hindi text responses are converted into natural-sounding speech and played through the speaker.

Audio Output

The synthesized speech is delivered through speakers connected via the 3.5 mm audio jack or HDMI.

4. METHODOLOGY

The development of the offline Hindi voice assistant was carried out in a sequence of well-defined stages to ensure reliable operation and efficient performance on embedded hardware.

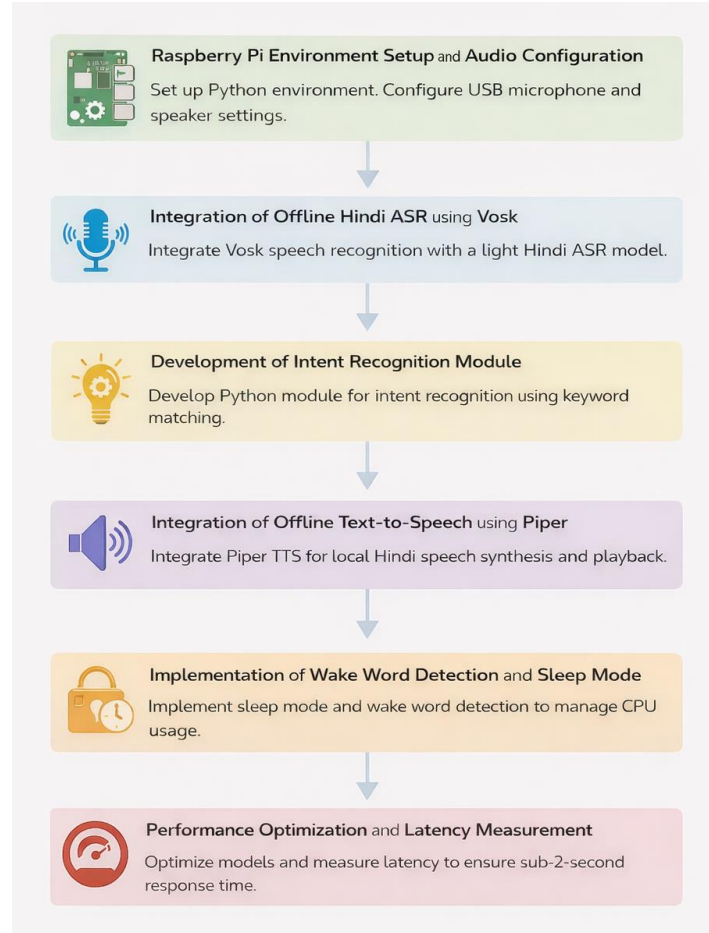


Figure 2: *Methodology flowchart*

Raspberry Pi Environment Setup and Audio Configuration

Raspberry Pi was configured by installing the necessary software and a Python environment. A virtual environment was used to avoid dependency conflicts.

Audio setup required a USB microphone to be connected for recording and speakers for playback. ALSA tools were used to list available devices, and the indices for recording and playing were determined from the list. Sampling rate and the number of channels were set according to the capabilities of the microphone to avoid recording errors or buffer overflows.

Integration of Offline Hindi ASR using Vosk

The offline speech recognition used a Vosk engine with a light-weight Hindi model integrated into a Python application. The microphone audio was streamed to this engine, recognizing speech in Hindi. The application then processed the output to filter out noise and false recognitions.

Development of Intent Recognition Module

Once the Hindi text is recognized, a Python module for intent recognition is developed. The Python module uses the keyword matching method for the predefined commands like 'time,' 'date,' 'greetings,' 'music control,' and 'volume control,' as it is a lightweight and fast method.

Integration of Offline Text-to-Speech using Piper

To generate spoken responses, the Piper text-to-speech engine was integrated. A Hindi voice model was loaded locally to synthesize speech without internet connectivity. The generated audio was saved temporarily and played through the speaker using the sound output interface. The TTS model was preloaded at startup to reduce response delay during runtime.

Implementation of Wake Word Detection and Sleep Mode

Sleep mode has been introduced, which reduces the CPU's utilization and prevents any extra processing. If there is no command received for a certain period of time, the system will be in a low-activity listening state and will be fully active only when the predefined word is recognized.

Performance Optimization and Latency Measurement

Several optimizations were applied to improve system performance:

- Lightweight ASR and TTS models were used
- Audio buffer sizes were tuned to reduce overflow errors
- TTS models were preloaded to reduce startup delay
- Latency measurement was implemented to evaluate response time

These optimizations ensured that the system met the requirement of responding within two seconds while operating completely offline.

5. HARDWARE UTILIZED

The following hardware components were used to implement the offline Hindi voice assistant:

RASPBERRY PI 4 MODEL B

The Raspberry Pi serves as the main processing unit of the system. All speech recognition, intent detection, and speech synthesis are performed locally on this device without relying on external servers.



Figure 3: *Raspberry Pi 4 Model B*

USB Microphone

A USB microphone is used to capture the user's voice commands. It provides real-time audio input to the speech recognition module.

Speaker / Headphones

Speakers or headphones connected through the audio jack or HDMI are used to play the synthesized speech responses generated by the assistant.

Micro SD Card & SD Card Reader

The MicroSD card is used to store the Raspberry Pi operating system, project files, speech models, and dependencies required for offline operation.

6. SOFTWARE UTILIZED

The following software tools and libraries were used in the development of the system:

Raspberry Pi OS

The operating system used to run the application and manage hardware resources on the Raspberry Pi.

Python 3

Python was used as the primary programming language for implementing speech processing, intent recognition, and system control logic.

Vosk Speech Recognition Toolkit

Vosk was used as the offline Automatic Speech Recognition (ASR) engine to convert Hindi speech into text locally on the Raspberry Pi.

Piper Text-to-Speech (TTS)

Piper was used to convert response text into spoken Hindi audio without requiring an internet connection.

SoundDevice and SoundFile Libraries

These libraries were used for capturing audio input and playing synthesized speech output.

ALSA (Advanced Linux Sound Architecture)

ALSA was used for configuring and managing audio devices such as the microphone and speaker.

MPG123

This utility was used for playing audio files such as music during command execution.

7. OPTIMIZATION TECHNIQUES

To ensure smooth operation on resource-constrained embedded hardware and to meet the low-latency requirement, several optimization techniques were implemented during the development of the offline Hindi voice assistant.

Use of Lightweight Speech Models

A lightweight Hindi speech recognition model was selected for the ASR module to reduce processing time and memory usage. This helped achieve faster recognition while maintaining acceptable accuracy.

Keyword-Based Intent Recognition

Instead of using computationally intensive machine learning models for intent detection, a keyword-based approach was implemented. This method significantly reduced processing overhead and provided faster command recognition suitable for real-time applications.

Preloading of TTS Model

The Piper text-to-speech model was loaded once at system startup. This avoided repeated loading delays and reduced the time required to generate spoken responses.

Sleep Mode and Wake-Word Mechanism

To prevent unnecessary CPU usage, a sleep mode was implemented that activates after a period of inactivity. The system resumes full operation only when a predefined wake word is detected. This reduced processor load and helped control temperature during continuous operation.

Audio Buffer Optimization

Audio buffer sizes and sampling parameters were adjusted to minimize input overflow errors and ensure stable audio streaming during real-time speech recognition.

8. RESULTS AND PERFORMANCE EVALUATION

The offline Hindi voice assistant was evaluated on a Raspberry Pi device using various predefined Hindi voice commands like the display of time, date, greetings, music, and volume control. The offline Hindi voice assistant was able to detect audio input, translate speech to text using offline ASR, identify the correct action, and respond to the user using offline TTS.

During the evaluation process, the voice assistant was able to correctly identify predefined Hindi commands in real-time. The time taken to respond to the command was measured from the time the command was identified to the time audio playback began. The average response time was found to be around 1 to 2 seconds, which meets the performance criteria specified in the problem statement.

The voice assistant was also able to function correctly when continuously used. The addition of sleep mode and wake-word detection features enabled the CPU to work less and avoid processing unnecessary tasks when idle. Audio playback, speech recognition, and command execution were successfully completed without internet connectivity, thus ensuring that the entire process takes place offline.

PERFORMANCE SUMMARY

Parameter	Observation
Operating Mode	Fully Offline
Supported Commands	10–15 Hindi commands
Average Response Time	~1–2 seconds
Speech Recognition	Real-time, stable
System Stability	Stable during continuous operation

Table 1: *Performance summary*

DEMO VIDEO: A demonstration of the offline Hindi voice assistant responding to various Hindi commands can be viewed at the following link https://drive.google.com/file/d/1spGBemikwdPYMkIERIIAloKYTqv3Vz0c/view?usp=drive_link

SAMPLE TERMINAL OUTPUTS

Assistant Ready...

Assistant: तैयार हूँ
input overflow

You said: नमस्ते

Intent: GREETING

Assistant: नमस्ते

Response time: 0.578 sec

Total time: 1.794 sec

You said: परिचय

Intent: INTRO

Assistant: मैं रोही एक ऑफलाइन असिस्टेंट हूँ

Response time: 0.829 sec

Total time: 2.435 sec

You said: समय

Intent: GET_TIME

Assistant: अब समय 13:47

Response time: 0.098 sec

Total time: 2.459 sec

You said: दिनांक

Intent: GET_DATE

Assistant: तारीख 18-02-2026

Response time: 1.023 sec

Total time: 3.077 sec

You said: मदद

Intent: HELP

Assistant: समय या संगीत बोलें

input overflow

Response time: 1.205 sec

Total time: 2.871 sec

You said: बजाए

Intent: UNKNOWN

Assistant: समझ नहीं आया

Response time: 0.095 sec

input overflow

Total time: 2.01 sec

You said: ठीक है

Intent: STATUS

Assistant: सब ठीक है

input overflow

Response time: 0.643 sec

Total time: 1.794 sec

input overflow

input overflow

input overflow

You said: आपका धन्यवाद

Intent: THANKS

Assistant: स्वागत है

input overflow

Response time: 0.763 sec

Total time: 1.712 sec

input overflow

input overflow

You said: संगीत बजाओ

Intent: PLAY_MUSIC

Assistant: संगीत चला रहा हूँ

input overflow

Response time: 0.641 sec

Total time: 2.306 sec

High Performance MPEG 1.0/2.0/2.5 Player for Layers 1, 2 and 3
version 1.32.10; written and copyright by Micheal Mipp and others
free software (LPGL) without any warranty but with Best wishes

input overflow

Terminal control enabled, Press 'A' for testing of Keys and functions.

Playing MPEG stream 1 of 1: song.mp3 ...

MPEG 1.0 L III vbr 44100 j-0

Title: Aavan Javan :: SenSongsMp3.Con

Artist: Arjith Singh, Nikhita Gandhi

Album: War 2 (2025)

Genre: Hindi

input overflow

You said: वॉल्यूम तेज़ करो

Intent: VOLUME_UP

Assistant: आवाज़ बढ़ा रहा हूँ

Response time: 0.639 sec

Simple mixer control 'PCM', 0

Capabilities: pvolume pvolume-joined pswitch pswitch-joined

Playback channels: Mono

Limits: Playback -10239 - 400

Mono: Playback -664 [90%] [-6.64dB] [on]

Total time: 2.176 sec

```

You said: वॉल्यूम कम करो
Intent: VOLUME_DOWN
Assistant: आवाज़ कम कर रहा हूँ
Response time: 0.069 sec
Simple mixer control 'PCM',0
  Capabilities: pvolume pvolume-joined pswitch pswitch-joined
  Playback channels: Mono
  Limits: Playback -10239 - 400
  Mono: Playback -1728 [80%] [-17.28dB] [on]
Total time: 2.343 sec
input overflow
input overflow
input overflow

You said: संगीत रोक
Intent: STOP_MUSIC
Assistant: संगीत बंद
Response time: 0.491 sec
input overflow

mpg123:death by SIGTERM
Total time: 1.745 sec

[0:24] Decoding of song.mp3 finished
input overflow

```

```

You said: बंद करो
Intent: EXIT
Assistant: स्लीप मोड
input overflow
input overflow

input overflow
input overflow
input overflow
input overflow
Sleep heard: सुनो असिस्टेंट
Assistant: तैयार हूँ
input overflow

```

9. CONCLUSION

In this project, a fully offline Hindi voice assistant was successfully developed and deployed on a Raspberry Pi. The system is capable of capturing voice input, recognizing Hindi speech locally, detecting user commands, and generating spoken responses without relying on internet connectivity.

The assistant was tested with multiple predefined Hindi commands and demonstrated reliable performance with response times within the required limits. Optimization techniques such as lightweight models, sleep mode, and wake-word detection helped improve efficiency and maintain stable operation on embedded hardware.

This project demonstrates that voice-based human-computer interaction can be effectively implemented on resource-constrained ARM-based systems while ensuring privacy, low latency, and offline operation. Future improvements may include supporting additional commands, integrating IoT devices, and further optimizing performance.

10. FUTURE SCOPE

The current system can be further improved by increasing the number of supported Hindi commands and enhancing speech recognition accuracy for different accents. Integration with IoT devices can enable the assistant to control home appliances and perform additional tasks.

In the future, the assistant can be further automated to run more efficiently at system startup and operate as a fully standalone device without manual intervention. Additional improvements may include support for multiple Indian languages and further optimization to reduce response time and power consumption on embedded hardware.

11. REFERENCES

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3. Raspberry Pi Documentation – <https://www.raspberrypi.com/documentation/>
4. SoundDevice Python Library Documentation – <https://python-sounddevice.readthedocs.io/>
5. ALSA (Advanced Linux Sound Architecture) Documentation – <https://www.alsa-project.org/>

PROJECT GITHUB LINK: <https://github.com/jayanth-55/offline-hindi-voice-assistant>